SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence **COURSE CODE: CSA1741**

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COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively.*(BL3)*

Assessment Tool Weightage: 10%

QUESTIONS:2 **Total Marks:20**

Descriptive Test

1. A smart home automation system is being designed to manage lighting, temperature, and security based on user behaviour and environmental conditions. The system must respond to real-time inputs such as motion detection, temperature changes, and time of day while also learning user preferences over time. In this context, explain the different types of intelligent agents (simple reflex, model-based, goal-based, and utility-based agents) and analyze how each type would function within this system. Justify which type of agent or hybrid approach would be most effective for ensuring comfort, energy efficiency, and security.
2. A robot is placed in an unknown maze and must find a path to the exit without prior knowledge of the layout. Explain how uninformed search strategies like Uniform Cost Search (UCS) and Iterative Deepening Search (IDS) can help solve this problem. Discuss the advantages and disadvantages of these algorithms in terms of time and space complexity. Relate the problem to different types of intelligent agents and explain how agent design affects decision-making. Suggest which combination of agent type and search strategy would be most effective and justify your choice.

Code of Conduct:

*I **Brahmendra babu(192312439)** certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Brahmendra

RUBRICS FOR CALCULATION:

Criterion	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Max Marks
Criteria	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**	**Max Marks**
Understanding of Concepts	Demonstrates thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions	5
Explanation	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation	5
Application	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis	4
Organization	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers	3
Accuracy	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps	2
Clarity	Clear, neat, professional presentation	Understandable with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand	1

QUESTION 1: INTELLIGENT AGENTS IN SMART HOME AUTOMATION

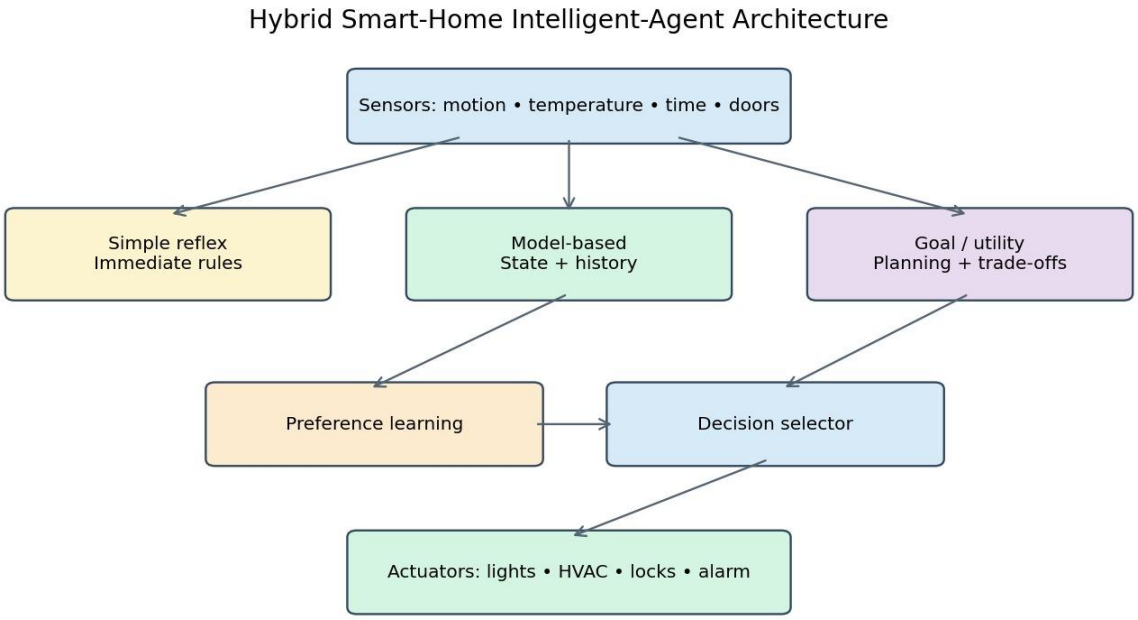
1.1 Introduction

A smart home automation system manages lighting, temperature, and security using sensors, software, and actuators. It receives real-time inputs such as motion detection, temperature changes, time of day, door status, and user actions. An intelligent agent observes these inputs and selects actions to improve comfort, energy efficiency, convenience, and safety. A learning component can adapt non-critical settings to user preferences over time.

1.2 Basic intelligent-agent model

An intelligent agent follows a perception–decision–action cycle. Sensors provide percepts; the agent interprets the current situation and chooses an action; actuators perform the action. Feedback helps the system update its internal state.

Component	Smart-home example
Sensors / percepts	Motion detector, thermostat, light sensor, clock, door/window sensor
Agent program	Rules, internal state, goals, utility function, preference model
Actuators	Lights, HVAC, smart locks, alarm, blinds
Performance measures	Comfort, energy use, response time, security, user satisfaction



1.3 Types of intelligent agents

A. Simple reflex agent

A simple reflex agent selects an action using condition–action rules based only on the current percept. It does not maintain a detailed history. Example: IF motion is detected AND it is dark, THEN switch on the hallway light. IF smoke is detected, THEN activate the alarm.
Advantages: fast, simple, and suitable for urgent, clearly defined responses.
Limitations: cannot reason well about hidden conditions or past events; fluctuating sensor readings may cause repeated switching.
Use: motion lighting, smoke alarms, immediate intrusion alerts.

B. Model-based agent

A model-based agent maintains an internal state based on current percepts and previous observations. It may track whether a room is occupied, whether a door was opened, current temperature, and the state of lights or HVAC.
Advantages: handles partial observability and reduces inconsistent decisions by using history.
Limitations: requires a reliable model; stored information can become outdated.
Use: occupancy-aware lighting, door-state tracking, avoiding unnecessary HVAC changes.

C. Goal-based agent
A goal-based agent evaluates actions according to whether they help achieve a desired state. Goals may include reaching a target temperature, turning off lights in empty rooms, or ensuring doors are locked when occupants leave.
Advantages: supports planning and can handle different routines such as sleep, away, and arrival modes.
Limitations: a goal alone may not distinguish between two successful actions with different energy costs or comfort levels.

D. Utility-based agent
A utility-based agent assigns numerical values to possible outcomes and selects the action with the best utility. It is useful when objectives conflict, such as higher comfort versus lower energy consumption.
Advantages: compares trade-offs and selects among several acceptable actions.
Limitations: utility scores and weights must be selected carefully; security should not be sacrificed for comfort.

1.4 Comparison

Agent type	Information used	Example	Limitation
Simple reflex	Current input	Motion + darkness turns on light	No history
Model-based	Current input + internal state	Tracks room occupancy	Model can become stale
Goal-based	State + desired goal	Activate away mode	Goals may not compare costs
Utility-based	State + utility/preferences	Balance comfort and energy	Depends on appropriate weights

1.5 Worked utility calculation
Assume two heating actions are scored from 0 to 10. These numbers and weights are illustrative, not measurements supplied in the assessment. Let:
U = 0.6 × Comfort score + 0.4 × Energy-efficiency score

Action	Comfort	Energy score	Calculation	Utility
A: Higher heating	9	4	0.6(9) + 0.4(4) = 5.4 + 1.6	7.0
B: Moderate heating	7	8	0.6(7) + 0.4(8) = 4.2 + 3.2	7.4

Action B has utility 7.4, which is higher than Action A's 7.0, so the utility-based agent selects moderate heating. If comfort is more important to a user, the weights can be adjusted. Safety-

critical rules, such as preventing unauthorized unlocking, should be hard constraints rather than trade-offs.

1.6 Real-time operation and learning

Collect real-time sensor inputs such as motion, temperature, time, and door status.
Update the internal state, including occupancy estimates and device status.
Apply immediate reflex rules to emergencies such as smoke or intrusion.
Evaluate goals and candidate actions; use utility scores to balance comfort and energy.
Learn preferences from repeated user adjustments, while allowing manual override.
Monitor the result and update the state after the actuators operate.

1.7 Recommended design: hybrid agent

A hybrid design is most effective. Simple reflex rules provide fast emergency responses; a model-based layer tracks occupancy and device state; a goal-based planner manages routines; and a utility-based layer balances comfort, convenience, and energy. Preference learning personalizes non-critical choices.

Requirement	Mechanism	Reason
Fast response	Simple reflex rules	Low latency for alarms and motion lights
Partial observability	Model-based state	Uses history to infer occupancy and device state
Schedules and routines	Goal-based planning	Supports sleep, away, and arrival modes
Comfort vs energy	Utility-based selection	Compares competing outcomes
Security	Hard safety rules	Safety should not be overridden by utility
Personalization	Preference learning	Adapts routine settings

1.8 Conclusion

Simple reflex agents are fast, model-based agents maintain state, goal-based agents plan, and utility-based agents compare trade-offs. A hybrid system with explicit security constraints combines these strengths and best supports comfort, energy efficiency, and security.

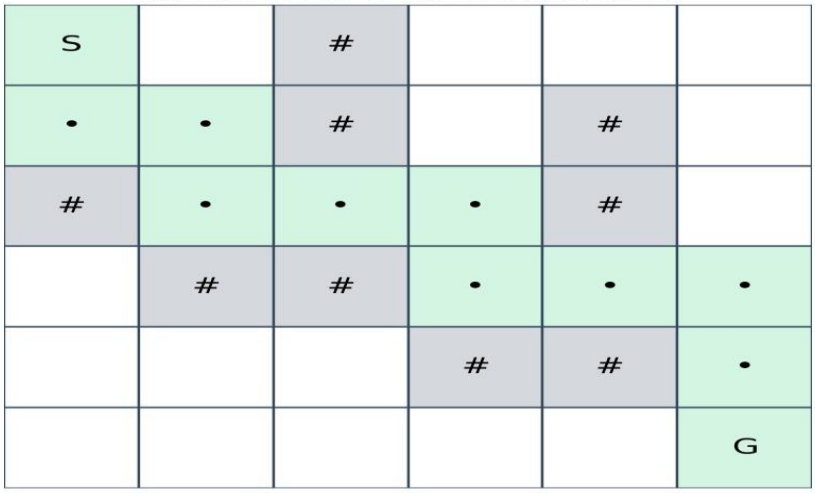
QUESTION 2: SOLVING AN UNKNOWN MAZE USING UCS AND IDS

2.1 Problem statement and assumptions

A robot is placed in an unknown maze and must reach the exit without knowing the complete layout. It senses nearby cells, discovers walls and corridors, and updates its internal map. The question asks how Uniform Cost Search (UCS) and Iterative Deepening Search (IDS) help, their time and space complexity, how agent type affects decisions, and which combination is most effective.

The assessment does not provide exact maze coordinates or movement costs. The maze and numerical route below are illustrative examples, not unique answers derived from the source.

Illustrative Maze and Example Route

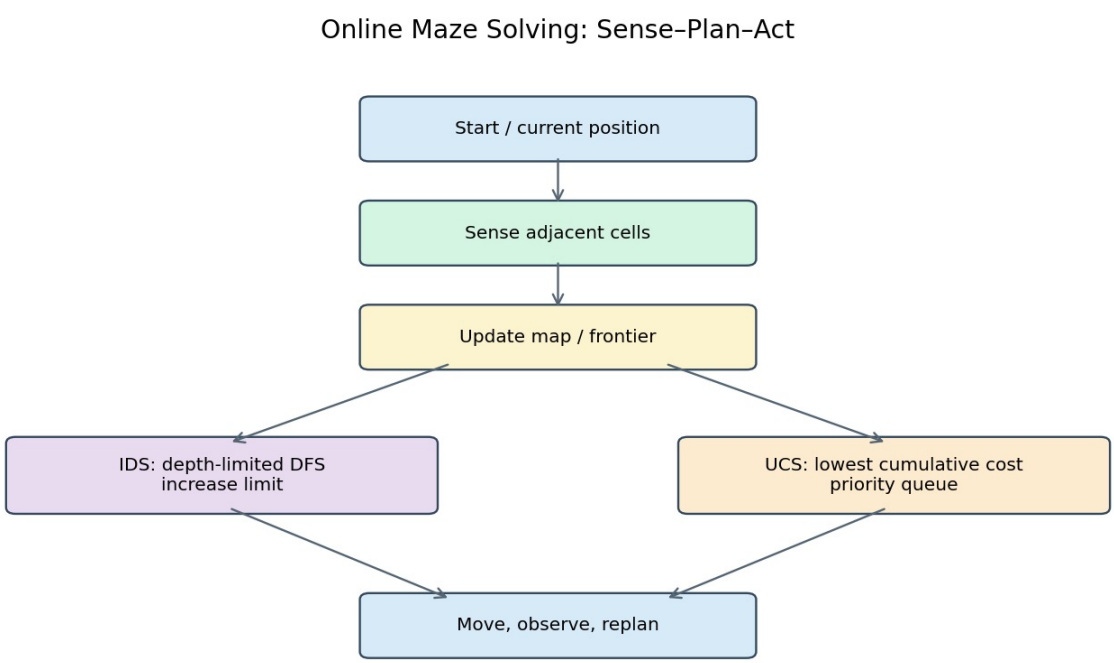


2.2 Uniform Cost Search (UCS)

1. UCS expands the frontier node with the lowest accumulated path cost g(n), using a priority queue. When every step has the same cost, UCS behaves similarly to BFS. When step costs differ but are non-negative, UCS finds a minimum-cost path.
2. Place the start node in the priority queue with cost 0.
3. Remove the node with the smallest cumulative cost.
4. If it is the exit, reconstruct and return the path.
5. For each legal neighbor, calculate g(new) = g(current) + step cost.
6. If this route is cheaper than the previously recorded route, update the cost and parent.
7. Continue until the exit is reached or the queue is empty.

2.3 Iterative Deepening Search (IDS)

1. IDS repeatedly runs depth-limited depth-first search with increasing depth limits: 0, 1, 2, 3, and so on. If the exit is not found within the current limit, the limit is increased and the search restarts. IDS combines low memory use with the ability to find a shallowest solution when all actions have equal cost.
2. Set depth limit L = 0.
3. Run depth-limited DFS from the start, stopping expansion at depth L.
4. If the exit is found, return the path.
5. Otherwise increase L by 1 and repeat.
6. Track cells on the current path or otherwise handle cycles to avoid endless looping.



2.8 Recommended combination

Maze condition	Recommended approach	Justification
Equal step costs and limited memory	Model-based goal agent + IDS	IDS uses little memory and finds a shallowest solution.
Different non-negative movement costs	Model-based goal agent + UCS	UCS minimizes total route cost rather than only number of moves.
Maze revealed during movement	Online model-based agent + repeated planning	Uses local sensing to update the map and replan.

If the robot must minimize total cost and movement costs differ, UCS with a model-based, goal-driven agent is the best choice among the two strategies. If all steps cost the same and memory is the main constraint, IDS is a good alternative. Since the maze is unknown, the robot also needs sensing, memory, and replanning; UCS or IDS alone cannot reveal unseen passages.

2.9 Conclusion

UCS is appropriate for different non-negative movement costs because it expands the lowest-cost frontier and finds a least-cost path. IDS is useful when all actions cost the same and memory is limited. A model-based goal-driven agent is effective in an unknown maze because it records observations, searches toward the exit, avoids known dead ends, and replans as the map becomes clearer.